Superstore Cloud Analytics Dashboard

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PROJECT OVERVIEW

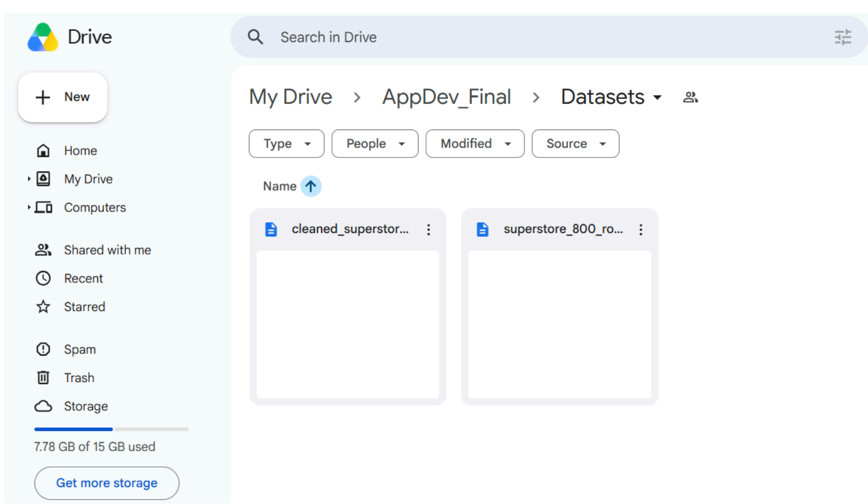
This project is a cloud-based Big Data analytics dashboard designed to process, analyze, and visualize Superstore sales data. The system begins by cleaning and transforming the raw dataset using Python and Pandas, including handling missing values, normalizing sales data, and removing outliers. The cleaned dataset is then stored in an SQLite database where analytical queries are performed to extract meaningful business insights such as total sales per category, average sales per region, and top-performing products.

The project also includes an interactive web dashboard developed using HTML, CSS, JavaScript, and Chart.js. The dashboard displays KPI cards and different chart visualizations that help users better understand sales performance and trends. In addition, the system integrates an AI-powered insight generator using the OpenRouter API, which analyzes real dataset information and produces business recommendations automatically.

DATASET DESCRIPTION

The dataset used in this project is a Superstore retail sales dataset containing 682 cleaned transaction records after preprocessing and outlier removal. The dataset includes important fields such as Order ID, Region, Category, Sub-Category, Product Name, and Sales. These columns provide meaningful business information that can be analyzed to understand sales performance, customer purchasing behavior, and product trends across different regions.

The dataset was selected because it represents real-world retail business operations and is suitable for Big Data analytics tasks such as data cleaning, aggregation, querying, and visualization. The data was first reduced to a valid range based on the project requirements, then cleaned and transformed using Python and Pandas before being stored and analyzed in the system.



SYSTEM ARCHITECTURE

Raw CSV Dataset
↓
Python Cleaning Script
↓
Cleaned CSV Dataset
↓
SQLite Database
↓
Analytical Queries
↓
HTML/CSS/JavaScript Dashboard
↓
OpenRouter AI Insight Generation

The system architecture shows the complete data flow of the Superstore Cloud Analytics Dashboard. The process starts with the raw Superstore CSV dataset, which is cleaned and transformed using Python and Pandas. During this stage, missing values are handled, sales values are normalized, and outliers are removed to produce a cleaner dataset.

After preprocessing, the cleaned dataset is loaded into an SQLite database where the records are stored in a structured table. Analytical queries are then performed to answer business questions such as total sales per category, average sales per region, and top-performing products. The results are used to support the web-based dashboard, which was created using HTML, CSS, JavaScript, and Chart.js. The system also includes AI integration through OpenRouter, which uses real dataset summary values to generate business insights and recommendations.

DATA CLEANING AND TRANSFORMATION

clean.py > ...

```python
import pandas as pd

df = pd.read_csv("superstore_800_rows.csv")

df = df.dropna()

df["sales_norm"] = (df["Sales"] - df["Sales"].min()) / (df["Sales"].max() - df["Sales"].min())

Q1 = df["Sales"].quantile(0.25)
Q3 = df["Sales"].quantile(0.75)
IQR = Q3 - Q1

df = df[(df["Sales"] >= Q1 - 1.5*IQR) & (df["Sales"] <= Q3 + 1.5*IQR)]

df.to_csv("cleaned_superstore.csv", index=False)

print("DONE CLEANING")
```

```
PS C:\Users\63932\Downloads\SUPERSTORE_PROJECT> python clean.py
DONE CLEANING
```

The data cleaning and transformation process was performed using Python and the Pandas library. The raw dataset was first checked for missing or null values, and incomplete records were removed using the dropna() function to improve data consistency and accuracy.

After handling missing values, the Sales column was normalized using the min-max scaling method in order to standardize numerical values for analysis. Outliers were then detected and removed using the Interquartile Range (IQR) method to reduce the effect of extreme values on the dataset. Once preprocessing was completed, the cleaned dataset was exported into a new CSV file named cleaned_superstore.csv for use in the database, analytical queries, dashboard visualization, and AI insight generation.

DATABASE DESIGN AND LOADING

load_database.py > ...

```python
import pandas as pd
import sqlite3

# Load the cleaned dataset
df = pd.read_csv("cleaned_superstore.csv")

# Create/connect to SQLite database
conn = sqlite3.connect("superstore.db")

# Load data into a table named superstore_sales
df.to_sql("superstore_sales", conn, if_exists="replace", index=False)

# Check if data loaded successfully
result = pd.read_sql_query("SELECT * FROM superstore_sales LIMIT 5", conn)
print(result)

# Show total rows
count = pd.read_sql_query("SELECT COUNT(*) AS total_rows FROM superstore_sales", conn)
print(count)

conn.close()

print("Database loaded successfully!")
```

```
PS C:\Users\63932\Downloads\SUPERSTORE_PROJECT> python load_database.py
   Row ID        Order ID  ...  Sales sales_norm
0     533  US-2018-129441  ...  47.94   0.005870
1     873  CA-2015-148488  ...  11.36   0.001297
2    1150  CA-2016-112452  ...  10.95   0.001246
3    2288  US-2018-112928  ...  17.48   0.002062
4    4039  CA-2015-110786  ...  21.12   0.002517

[5 rows x 19 columns]
   total_rows
0         682
Database loaded successfully!
```

The cleaned dataset was loaded into an SQLite database using Python and the sqlite3 library. A database named superstore.db was created to store the processed records in a structured table called superstore_sales. SQLite was selected because it is lightweight, free, and suitable for handling structured tabular datasets in analytics applications.

The load_database.py script was used to connect to the database, transfer the cleaned CSV dataset into the database table, and verify successful loading by displaying sample records and the total number of rows. This database setup allows the system to efficiently run analytical queries and support the dashboard visualizations.

ANALYTICAL QUERIES AND INSIGHTS

queries.py > ...

```python
import sqlite3
import pandas as pd

conn = sqlite3.connect("superstore.db")

# Query 1: Total sales per category
query1 = """
SELECT Category, ROUND(SUM(Sales), 2) AS total_sales
FROM superstore_sales
GROUP BY Category
ORDER BY total_sales DESC;
"""

# Query 2: Average sales per region
query2 = """
SELECT Region, ROUND(AVG(Sales), 2) AS average_sales
FROM superstore_sales
GROUP BY Region
ORDER BY average_sales DESC;
"""

# Query 3: Top 10 products by sales
query3 = """
SELECT "Product Name", ROUND(SUM(Sales), 2) AS total_sales
FROM superstore_sales
GROUP BY "Product Name"
ORDER BY total_sales DESC
LIMIT 10;
"""

print("\nQUERY 1: Total Sales per Category")
print(pd.read_sql_query(query1, conn))

print("\nQUERY 2: Average Sales per Region")
print(pd.read_sql_query(query2, conn))

print("\nQUERY 3: Top 10 Products by Sales")
print(pd.read_sql_query(query3, conn))

conn.close()
```

```
PS C:\Users\63932\Downloads\SUPERSTORE_PROJECT> python queries.py

QUERY 1: Total Sales per Category
          Category  total_sales
0  Office Supplies     26678.97
1        Furniture     16030.50
2       Technology     13817.75

QUERY 2: Average Sales per Region
    Region  average_sales
0     West          90.66
1    South          88.45
2  Central          83.34
3     East          69.58

QUERY 3: Top 10 Products by Sales
                                        Product Name  total_sales
0              Padded Folding Chairs, Black, 4/Carton       728.82
1                  Adjustable Depth Letter/Legal Cart       725.84
2                  Advantus Rolling Drawer Organizers       692.64
3          Situations Contoured Folding Chairs, 4/Set       631.72
4    Nortel Meridian M3904 Professional Digital phone       615.96
5   Deflect-o RollaMat Studded, Beveled Mat for Me...       608.72
6     Global Wood Trimmed Manager's Task Chair, Khaki       554.98
7               SAFCO Commercial Wire Shelving, Black       552.56
8                           Global Deluxe Steno Chair       515.77
9                 SanDisk Cruzer 64 GB USB Flash Drive       493.95
PS C:\Users\63932\Downloads\SUPERSTORE_PROJECT>
```

Three analytical queries were performed on the cleaned dataset stored in the SQLite database in order to extract meaningful business insights. These queries were created using SQL and executed through Python.

1. Total Sales per Category

This query calculates the total sales generated by each product category. The results help identify which category contributes the highest revenue to the business and which categories may require improvement in sales performance.

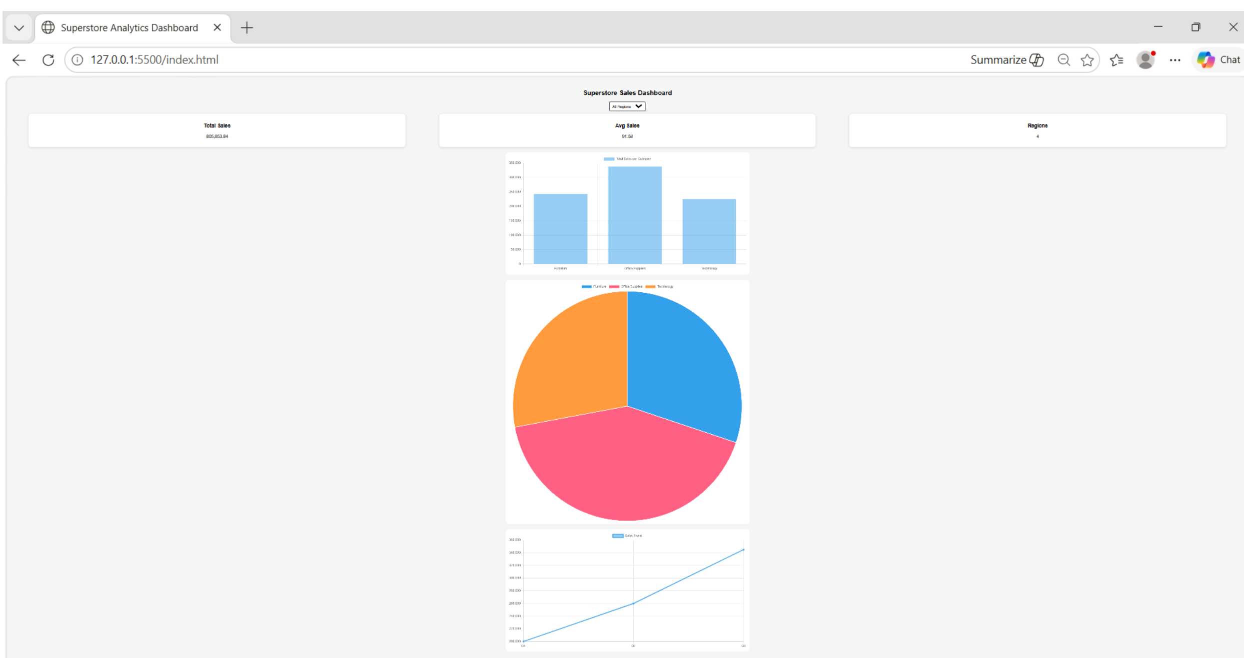
2. Average Sales per Region

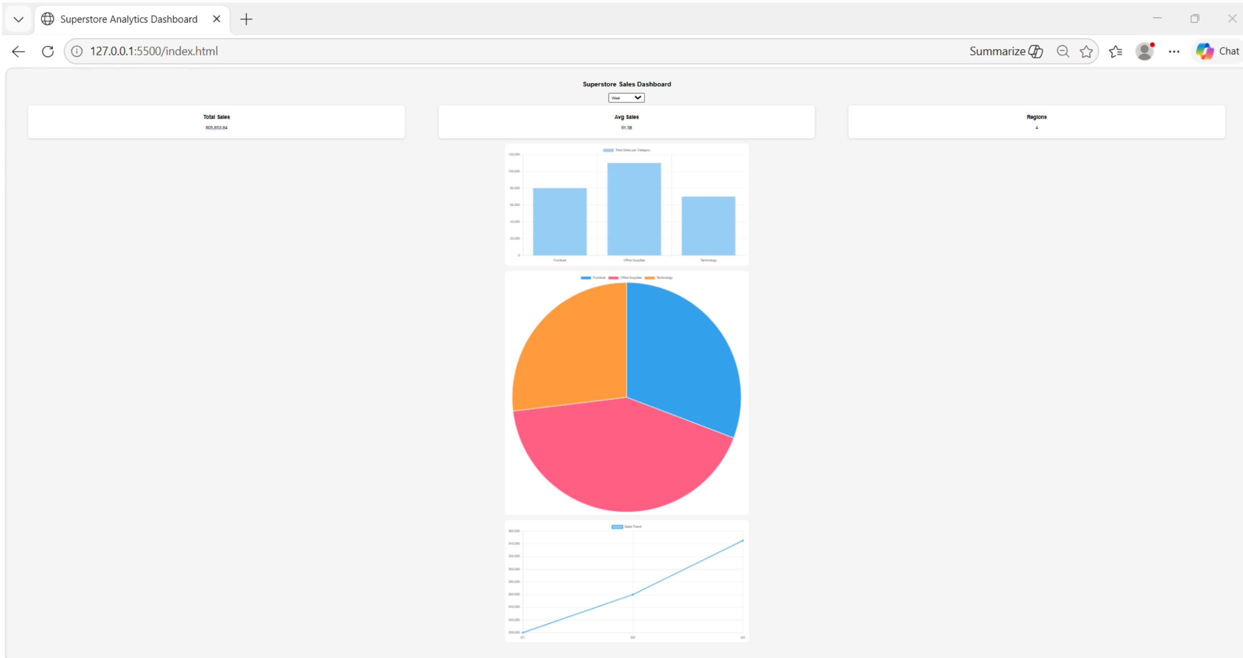
This query computes the average sales value for each region. The analysis allows comparison of regional sales performance and helps determine which geographic areas perform better in terms of customer purchases.

3. Top 10 Products by Sales

This query identifies the ten products with the highest total sales. The results help determine which products are the strongest contributors to business revenue and can support inventory and marketing decisions.

DASHBOARD IMPLEMENTATION





The analytics dashboard was developed using HTML, CSS, JavaScript, and Chart.js to provide an interactive visualization of the Superstore dataset. The dashboard displays important business information using multiple chart types including a bar chart, pie chart, and line chart. These visualizations help users better understand sales distribution, category performance, and sales trends.

The dashboard also includes KPI cards that display summary statistics such as total sales, average sales, and the total number of regions. A dropdown filter was implemented to allow users to interact with the dashboard and view sales data based on selected regions. The responsive layout allows the dashboard to be viewed on different screen sizes while maintaining readability and usability.

AI INTEGRATION

ai_insight.py > ...

```python
import requests
import pandas as pd

# Paste your OpenRouter API key here
API_KEY = [redacted]

# Load cleaned dataset
df = pd.read_csv("cleaned_superstore.csv")

# Create real dataset summary
total_sales = round(df["Sales"].sum(), 2)
average_sales = round(df["Sales"].mean(), 2)
top_category = df.groupby("Category")["Sales"].sum().idxmax()
top_region = df.groupby("Region")["Sales"].sum().idxmax()

# AI prompt
prompt = f"""
You are a business data analyst.

Analyze this retail dataset:

Total Sales: {total_sales}
Average Sales: {average_sales}
Top Category: {top_category}
Top Region: {top_region}

Write a short business insight and one recommendation.
"""

# Send request to OpenRouter AI
response = requests.post(
    url="https://openrouter.ai/api/v1/chat/completions",
    headers={
        "Authorization": f"Bearer {API_KEY}",
        "Content-Type": "application/json"
    },
    json={
        "model": "openrouter/free",
        "messages": [
            {
                "role": "user",
                "content": prompt
            }
        ]
    }
)

# Convert response to JSON
result = response.json()

print("\nFULL API RESPONSE:\n")
print(result)

# Check if AI response exists
if "choices" in result:
    print("\nAI GENERATED INSIGHT:\n")
    print(result["choices"][0]["message"]["content"])
else:
    print("\nERROR FROM API:")
    print(result)
```

```
AI GENERATED INSIGHT:

**Business Insight:**
Our retail dataset indicates a strong performance in total sales, with an average of 82.88
achieved across the shop. The top-selling category, Office Supplies, suggests a solid deman
d in this area, while an unnamed top region contributes significantly to revenue.

**Recommendation:**
To enhance profitability, consider expanding the product range in the top-performing top re
gion and optimizing marketing efforts specifically in the high-growth office supplies segme
nt. This could help capture emerging market trends and further boost sales.
```

The project includes an AI-powered insight generation feature integrated using the OpenRouter API and a free large language model. The AI system analyzes real values from the cleaned Superstore dataset such as total sales, average sales, top-performing category, and top-performing region. These values are automatically included in a prompt sent to the AI model.

The AI integration was implemented using Python and the Requests library. After processing the dataset summary, the AI generates a short business insight and recommendation based on the sales data. This feature demonstrates how artificial intelligence can assist businesses in understanding sales performance and making data-driven decisions automatically.

CHALLENGES ENCOUNTERED

Several challenges were encountered during the development process:

- File path errors when running Python scripts
- Missing file extensions causing execution issues
- Data cleaning required handling inconsistent or missing values
- Integrating dynamic charts required proper dataset structuring
- Reducing the original dataset to meet the required 500–1000 row limit

KEY LEARNINGS

This project provided valuable learning experiences in:

- Data preprocessing using Python and Pandas
- Analytics
- Working with real-world datasets
- Cloud-based data handling concepts
- Building interactive dashboards
- CMD py
- Understanding the full data pipeline from ingestion to visualization

Isagani Nermal

This project helped me improve my understanding of data analytics and dashboard development. I learned how to preprocess datasets using Python and how data visualization can support business decision-making.

Michael Angelo Balmeo

Through this project, I gained experience in using SQLite databases and analytical queries. I also learned how cloud-based analytics systems process and visualize real-world data.

John Luis Cofreros

This project improved my skills in Python scripting, database handling, and AI integration. I learned how APIs such as OpenRouter can be used to generate business insights from real datasets.

Jasper Roa Siscar

I learned how to work with datasets and create interactive dashboard visualizations using HTML, CSS, JavaScript, and Chart.js. The project also taught me the importance of data cleaning and preprocessing before analysis.

Chauncey Hamilton Villegas

This project helped me understand the complete workflow of a cloud analytics application from data ingestion to AI-generated insights. I also improved my teamwork and problem-solving skills during the development process.

CLOUD TECHNOLOGIES AND FREE TOOLS USED

Several free cloud technologies and development tools were used to build the Superstore Cloud Analytics Dashboard. Google Drive was used as the cloud storage platform for storing the raw and cleaned datasets. Python and the Pandas library were used for data preprocessing, normalization, outlier removal, database loading, and analytical query execution.

SQLite was selected as the database system because it is lightweight, free, and suitable for structured datasets. The dashboard interface was created using HTML, CSS, JavaScript, and Chart.js to provide interactive visualizations and KPI displays. OpenRouter was used as the free AI integration platform to generate business insights from real dataset values. Visual Studio Code served as the main development environment for coding and project management throughout the implementation process.